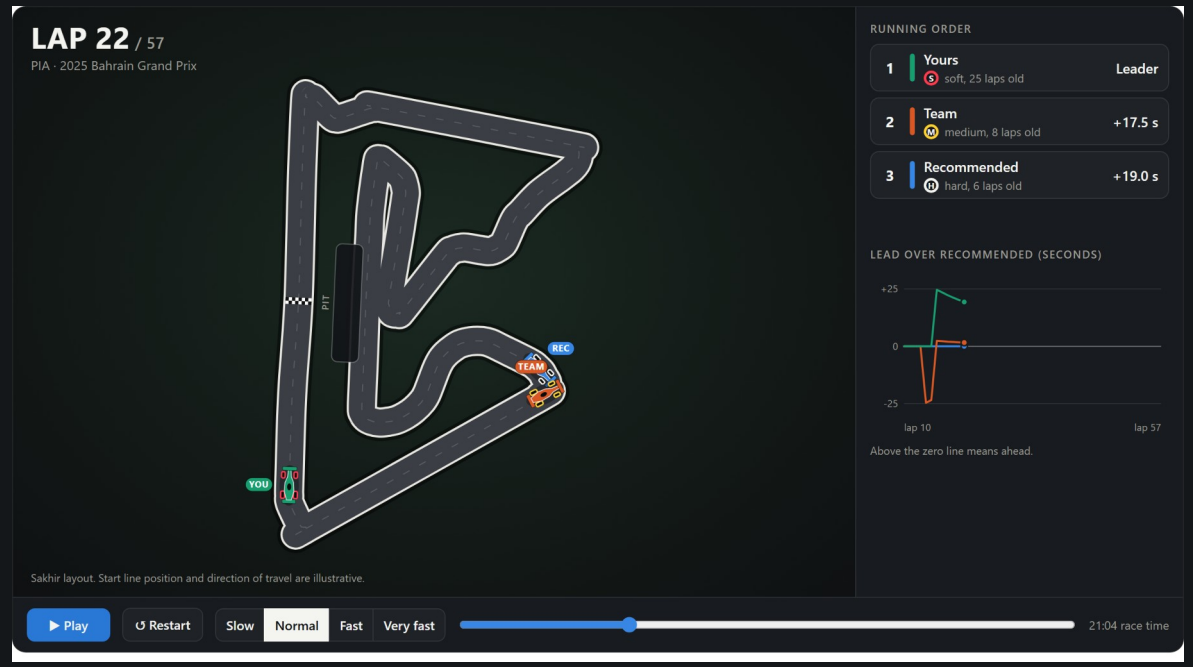
PREDICTIVE MODELLING · COURSE PROJECT

Can a model call the pit stop?

A statistical model of lap time, the optimisation built on it, and whether its error allows a decision

$$\hat{s} = \underset{s \in \mathcal{S}}{\arg\min} \; \hat{R}(s)$$



PROBLEM

Pit strategy as an optimisation over a learned function

OBJECTIVE

$$\hat{s} = \arg\min_{s \in \mathcal{S}} \hat{R}(s)$$

PREDICTED RACE TIME

$$\hat{R}(s) = \sum_{\ell=\ell_0}^{N} B\left(1 + \frac{\hat{f}(x_\ell(s))}{100}\right) + k(s)\,P$$

NOTATION

s	strategy: pit laps and tyre compounds
B	baseline pace of the driver, seconds
$\hat{f}$	learned model: % change in pace
$x_\ell(s)$	inputs on lap ℓ under strategy s
$k(s),\ P$	number of stops, time lost per stop
$\mathcal{S}$	feasible strategies (tyre rules)

1

Statistics: how accurate is the estimate?

Race time within 0.57% on an unseen season.

2

Decision: is the minimiser stable?

No. The recommended pit lap is 8 laps from the team's on average.

TARGET AND DATA

The response is a ratio, so the circuit and car scale cancel

RESPONSE

$$y_\ell = 100\left(\frac{T_\ell}{B} - 1\right)$$

BASELINE

$$B = \text{median}\{T_\ell : \ell \in \mathcal{B}\}, \qquad \mathcal{B} \subset \{2, \ldots, 12\}, \quad |\mathcal{B}| \leq 8$$

MODEL AND ITS INVERSE

$$y_\ell = f(x_\ell) + e_\ell \qquad \hat{T}_\ell = B\left(1 + \frac{\hat{f}(x_\ell)}{100}\right)$$

B is fixed a few laps into the race, so predicting later laps from it uses no future information.

$$x \in \mathbb{R}^6 \times \{\text{categories}\}^3$$

- Tyre age a (laps)
- Fuel burned since baseline / N
- Race progress ℓ / N
- Race length N
- Baseline tyre age
- Baseline track temperature
- Compound c: soft, medium, hard
- Baseline compound
- Circuit

Split	Seasons	Dry races	Laps n
Train	2022, 2023	35	24,039
Validation	2024	19	15,332
Test	2025	21	15,844

Split by season, not at random: errors within one race are correlated (ρ ≈ 0.33, slide 7), so a random split would leak.

Constant pace is biased; a linear model fixes most of it

NO CHANGE (PREDICT ZERO)

$$\mathrm{MSE} = \bar{y}^2 + s_y^2 : \qquad 1.698^2 = 1.181^2 + 1.220^2$$

Fuel burn makes cars faster: mean change 1.18% down. That bias gives $R^2 = -0.94$.

LINEAR REGRESSION (LEAST SQUARES)

$$\hat{\beta} = \arg\min_{\beta} \|y - X\beta\|^2 = (X^{\mathrm{T}}X)^{-1}X^{\mathrm{T}}y$$

13 coefficients. RMSE: training 0.913%, validation 0.930%. Almost no gap: low variance.

WEAR CURVE PER COMPOUND (RIDGE)

$$\min_{\beta} \|y - X\beta\|^2 + \alpha \|\beta\|^2, \qquad \alpha = 1$$

Adds a^2 and a × compound terms. Validation 0.938%: the extra terms do not help.

Validation season 2024

Model	RMSE %	RMSE s	R^2
Gradient boosting	0.902	0.798	0.454
Boosting, monotone wear	0.909	0.803	0.445
Linear regression	0.930	0.822	0.419
Linear, wear per compound	0.938	0.828	0.410
Mean change ($\hat{y} = \bar{y}$)	1.222	1.084	−0.002
No change ($\hat{y} = 0$)	1.698	1.550	−0.936

RMSE in % of baseline pace and in seconds per lap. Models were fitted on 2022–2023 and compared here; 2025 was not touched.

Boosting: gradient descent in function space

ADDITIVE MODEL

$$F_0(x) = \bar{y}, \qquad F_m(x) = F_{m-1}(x) + \nu\, h_m(x)$$

PSEUDO-RESIDUALS FOR SQUARED LOSS

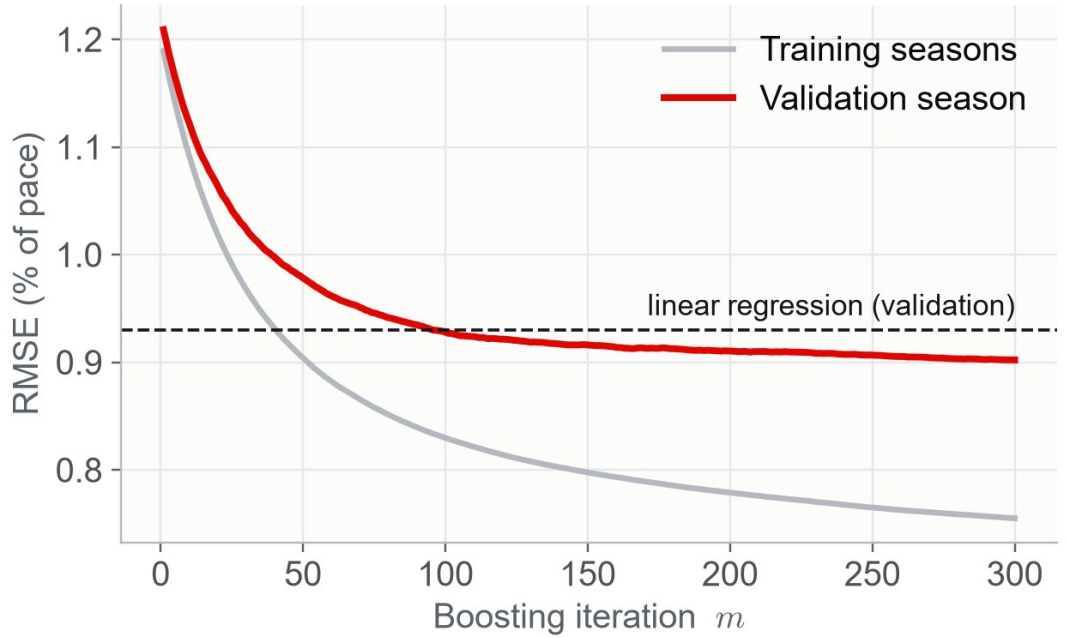
$$r_{im} = -\frac{\partial L}{\partial F}(x_i) = y_i - F_{m-1}(x_i), \qquad L = \tfrac{1}{2}(y - F)^2$$

LEAF VALUE

$$w_j = \frac{1}{n_j + \lambda} \sum_{i \in R_j} r_{im}$$

SPLIT GAIN

$$\frac{1}{2}\left[\frac{G_L^2}{n_L + \lambda} + \frac{G_R^2}{n_R + \lambda} - \frac{G^2}{n + \lambda}\right], \qquad G = \sum_i r_{im}$$



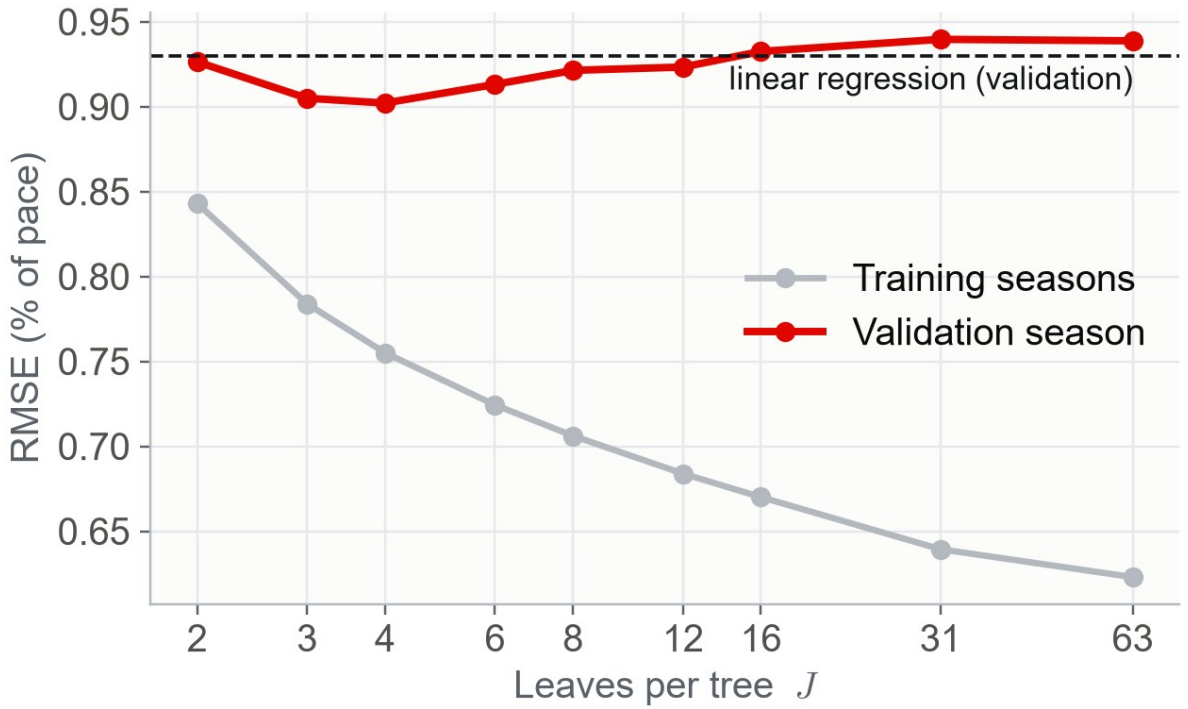
$M = 300$	$\nu = 0.03$	$J = 4$	$\lambda = 5$	$n_j \geq 300$

Trees, learning rate, leaves per tree, L2 penalty, minimum leaf size. The chart is the error after each tree.

MODEL SELECTION

Tree size is chosen by the bias–variance trade-off

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$



RMSE, % of baseline pace

Leaves J	Training	Validation	Gap
2	0.843	0.926	0.083
4	0.755	0.902	0.147
8	0.706	0.921	0.215
31	0.639	0.940	0.300
Linear	0.913	0.930	0.017

Training error falls with J; validation error is lowest at J = 4. The growing gap is variance.

Test season 2025, used once

0.835 s
RMSE per lap

1.558 s
constant pace

0.44
R^2

ERROR STRUCTURE

The errors are not independent: a three-level decomposition

VALIDATION RESIDUAL, BY LEVEL

$$e_{dj\ell} = a_d + b_{dj} + \varepsilon_{dj\ell}$$

$$a_d \sim (0, \sigma_a^2) \qquad b_{dj} \sim (0, \sigma_b^2) \qquad \varepsilon_{dj\ell} \sim (0, \sigma_\varepsilon^2)$$

d: driver-race, j: stint, ℓ: lap. Mutually independent, mean zero.

$\sigma_a = 0.545$	**Driver over a race** A misjudged baseline. The same for every strategy.
$\sigma_b = 0.438$	**Stint within a race** The part that separates strategies.
$\sigma_\varepsilon = 0.650$	**Lap within a stint** Averages out over a stint.

Share of error variance (in % of pace, squared)



INTRACLASS CORRELATION

$$\rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma_b^2 + \sigma_\varepsilon^2} = 0.33$$

EFFECTIVE SAMPLE SIZE

$$n_{\text{eff}} = \frac{n}{1 + (\bar{m} - 1)\,\rho} = \frac{15{,}332}{14.7} \approx 1{,}040$$

Propagating the error predicts the race-time error

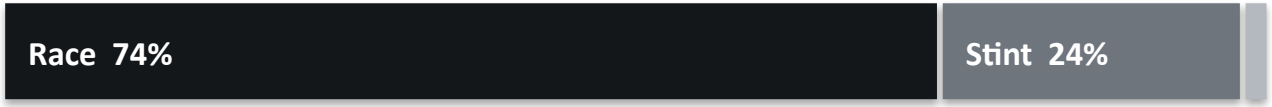
RACE-TIME ERROR FOR A GIVEN STRATEGY

$$R - \hat{R} = \frac{B}{100} \sum_{\ell=\ell_0}^{N} e_\ell$$

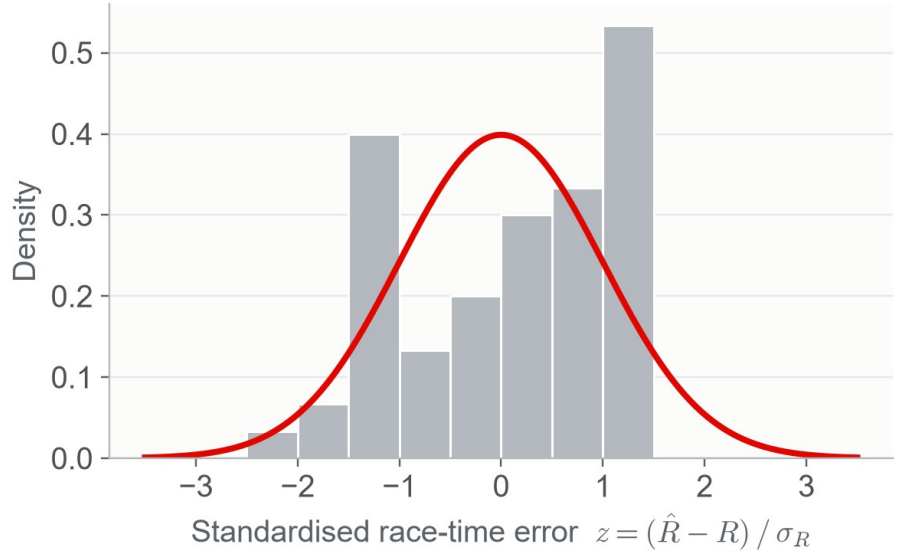
ITS VARIANCE

$$\sigma_R^2 = \left(\frac{B}{100}\right)^2 \left[N^2 \sigma_a^2 + \sigma_b^2 \sum_j n_j^2 + N \sigma_\varepsilon^2 \right]$$

Share of race-time variance (N = 50 laps, two stints of 25)



Lap noise is 46% of the lap-level variance but 2% of race-time variance: it averages out. The race-level term adds up coherently and dominates.



60 green-flag driver-races, 2025. Curve: standard normal. The spread matches; the shape is rough.

28.0 s
predicted SD, from 2024 variances

27.9 s
observed RMSE, 2025

1.03
SD of z (1 if the model holds)

OPTIMISATION

A small constrained search, solved exactly by enumeration

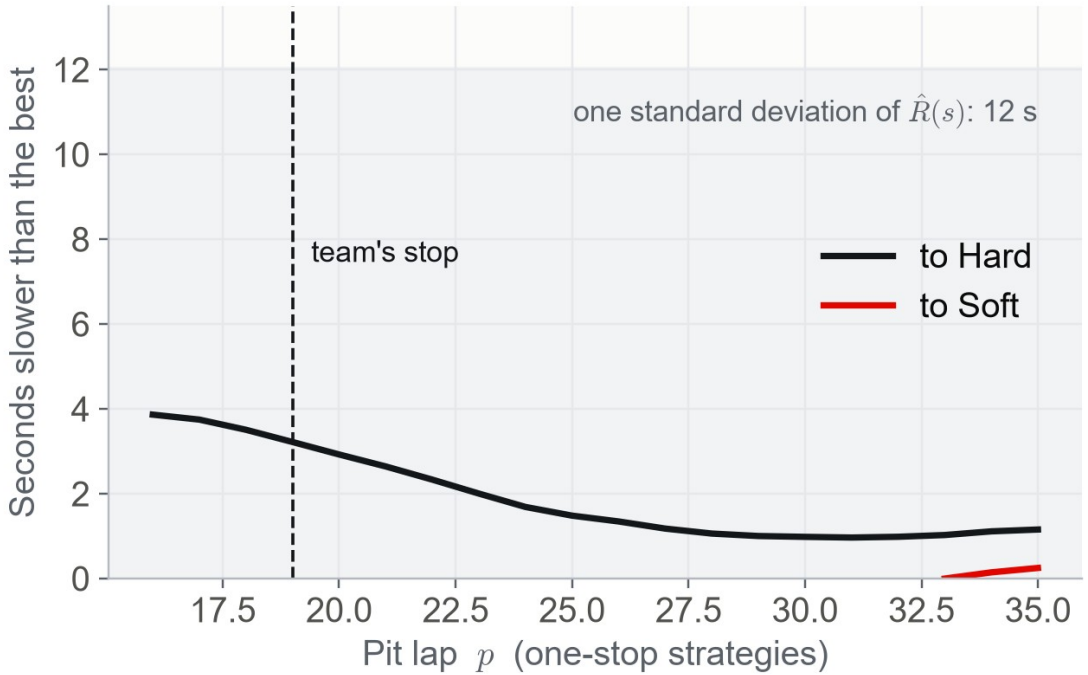
FEASIBLE SET

$$\mathcal{S} = \{\, s : \;\; n_j \geq 5, \quad |\{c_j\}| \geq 2, \quad a_j \leq A_{0.98}(c_j) \quad \forall j \,\}$$

n: stint length. c: compound. a: tyre age at the end of the stint.

Example: 2025 Saudi Arabian Grand Prix, winner

$3 \times 30 = 90$	one-stop candidates, 3 compounds
$9 \times 78 = 702$	two-stop candidates, pit-lap step 2
$\lvert\mathcal{S}\rvert = 497$	feasible: 23 one-stop, 474 two-stop
$497 \times 39 = 19{,}383$	lap predictions, one model call



The objective is flat near its minimum: 15 strategies are within 2 s of the best. The grey band is one standard deviation of a predicted race time (12 s).

Enumeration is exact, so the only error in the recommendation is the error in the predicted race time.

The minimiser is not stable: the signal is a fifth of the noise

NOISE SEPARATING TWO STRATEGIES (RACE TERM CANCELS)

$$\sigma^2(s) = \left(\frac{B}{100}\right)^2 \left[\sigma_b^2 \sum_j n_j^2 + N\sigma_\varepsilon^2\right]$$

PROBABILITY THE RECOMMENDATION IS REALLY FASTER

$$P = \Phi\left(\frac{\Delta}{\sigma_\Delta}\right), \qquad \Delta = \hat{R}(s') - \hat{R}(\hat{s}), \quad \sigma_\Delta^2 = \sigma^2(\hat{s}) + \sigma^2(s')$$

Example: 2025 Saudi Arabian Grand Prix, winner

$\Delta = 3.2$ s	predicted gain
$\sigma_\Delta = 18.2$ s	its standard deviation
$P = \Phi(0.18) = 0.57$	from the formula
$\hat{P} = 0.578 \pm 0.008$	Monte Carlo, M = 4000 races

All 182 driver-races of 2025: recommendation against the team's choice

65%
same number of stops

8.0 laps
mean distance of first pit lap

26%
within 3 laps

2.2 s
median predicted gain Δ

The standard deviation of Δ treats the two races as independent draws. Sharing the error on the laps both strategies have in common gives 16 s instead of 18 s.

DEMO

The same mathematics as an animated race



Circuit outlines are real. Start-line position and direction of travel are illustrative.

$\hat{s},\ s',\ s^{\text{you}}$

Three strategies race

The minimiser, the team's, and your own

$\hat{T}_\ell$

Every lap is a prediction

Baseline pace scaled by the model

$b_j,\ \varepsilon_\ell$

Random races

One Monte Carlo draw of stint and lap error

$\hat{R}(s)$

Live running order

Cumulative predicted time, gap to the leader

LIMITATIONS

Where the assumptions fail

1 Selection in tyre data

$$\mathbb{E}\,[\,y \mid a,\ \text{stint} \geq a\,] \ \leq\ \mathbb{E}\,[\,y \mid a\,]$$

Old tyres are seen only when they lasted. Stops come out later than the team's in 66% of cases, by 4.3 laps on average.

2 Omitted variables

$$y \ = \ f(x) \ + \ g(u) \ + \ \varepsilon, \qquad u \ \text{unobserved}$$

Traffic and track position are not in the data ($R^2 = 0.44$). Teams stop early to gain position; the model sees clean air only.

3 A regime that is not modelled

$$P \ \rightarrow \ P_{\text{SC}} \ < \ P \quad \text{under a safety car}$$

Safety cars make a stop cheaper and decide many races. Race time could be checked on 60 of 182 driver-races only.

4 Resolution of the decision

$$|\Delta| \approx 2\ \text{s} \ \ll \ \sigma_{\Delta} \approx 18\ \text{s}$$

Differences between good strategies are far inside the error. The team's choice is a reference, not a proven optimum.

Next: a survival model for stint length, a safety-car scenario with its own pit loss, and a traffic term.

CONCLUSION

An accurate model is not yet a decision

1	**Lap**	$\mathrm{RMSE} = 0.835\ \mathrm{s}, \quad R^2 = 0.44$	Unseen season. Constant pace: 1.558 s.
2	**Race**	$\sigma_R: \quad 28.0\ \mathrm{s\ predicted}, \quad 27.9\ \mathrm{s\ observed}$	The error model holds out of sample.
3	**Decision**	$\Delta / \sigma_\Delta \approx 0.2 \quad \Rightarrow \quad P \approx 0.57$	The minimiser is not identified.

Validate a model at the level of the decision it will be used for.